

Teorija polj
Vektorski operatorji

f = skalarno polje, $\mathbf{F}$ = vektorsko polje

grad: $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$

div: $\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

rot: $\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$

Laplacijan: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$

Lastnosti polj

Irotacijalno: $\nabla \times \mathbf{F} = 0$

Solenoidno: $\nabla \cdot \mathbf{F} = 0$

Konservativno: $\mathbf{F} = -\nabla \phi$

Potencial: $\phi = - \int \mathbf{F} \cdot d\mathbf{r}$

Prostorska geometrija
Koordinatni sistemi

Polarni: $x = r \cos \theta, \quad y = r \sin \theta$

Čilindrični: $x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$

Sferični: $x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \phi$

Dodatno

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)| \, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \, dt$

Tangentni vektor: $\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$

Normalna ravnina: $\mathbf{r}'(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$

Odvod integrala s parametrom:

$$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy \quad \Rightarrow$$

$$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy + f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$$

Naravna parameterizacija: $|\mathbf{r}'(s)| = 1, \quad s = \text{ločna dolžina}$

Večkratni integrali

Dvojni integral: $\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$

Ploščina: $A = \iint_D 1 \, dA$

Prostornina: $V = \iiint_D f(x, y) \, dA$

Težišče: $\bar{x} = \frac{1}{A} \iint_D x \, dA, \quad \bar{y} = \frac{1}{A} \iint_D y \, dA$

Trojni integral: $\iiint_E f(x, y, z) \, dV$

Masa: $m = \iiint_E \rho(x, y, z) \, dV$

Vztrajnostni moment: $I = \iiint_E r^2 \rho \, dV$

Povprečna vrednost: $\bar{f} = \frac{1}{V} \iiint_E f(x, y, z) \, dV$

Krivuljni integrali

Skalarno polje: $\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$

Vektorsko polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_a^b \mathbf{V}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$

Ploskovni integrali

Skalarno polje: $\iint_S f \, dS = \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} \, du \, dv$

kjer $E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$

Vektorsko polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] \, du \, dv$

Integralski izreki

Greenov: $\oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

Stokesov: $\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S}$

Gaussov: $\iint_S \mathbf{V} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{V} \, dV$

Kompleksna analiza
Elem. komp. fun.

$z = x + iy, \quad |z| = \sqrt{x^2 + y^2}, \quad \arg z = \arctan\left(\frac{y}{x}\right)$ (1)

$e^z = e^x (\cos y + i \sin y)$ (2)

$\log z = \ln |z| + i \arg z + 2\pi i k$ (3)

$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$ (4)

$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}$ (5)

Analitičnost in odvedljivost
Cauchy-Riemann equations:

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ (6)

Harmonic condition: $\nabla^2 u = \nabla^2 v = 0$
Complex Integration

Cauchy's formula: $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$ (7)

Residue theorem: $\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$ (8)

Laurent series: $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ (9)

Integrali
Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x)$

Po delih: $\int u \, dv = uv - \int v \, du$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

Odvodi
Pravila odvajanja

$$(f + g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Elementarni odvodi

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$